

Anytime-valid monitoring of technology adoption dynamics: Digital technology diffusion surveillance in the United States, 1990–2026

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Abstract

We develop an anytime-valid surveillance framework for technology adoption dynamics, combining a causal measurement layer that treats survey redesigns as interventions on the observation process with one-sided mixture e-processes whose error control holds uniformly over time. Applied to the U.S. Census Bureau’s biweekly AI-adoption survey, the monitor declares a diffusion takeoff on 20 April 2025, quantifies the mid-2025 deceleration as sub-threshold evidence later vindicated by re-acceleration, and correctly absorbs a 26-standard-error November 2025 question-wording redesign as measurement, unlike conventional repeated tests. A retrospective extension recovers three decades of internet and e-commerce diffusion shifts.

Keywords: technology diffusion, artificial intelligence adoption, anytime-valid inference, e-values, sequential monitoring, survey measurement, Business Trends and Outlook Survey

1. Introduction

How can a manager, analyst, or policymaker know—with stated statistical confidence, and at the moment it matters—that the diffusion of a new

12 technology has changed regime: that adoption has taken off, that growth
13 has stalled, or that an apparent shock is an artifact of how adoption is
14 measured? The question has become concrete with artificial intelligence.
15 Since September 2023 the U.S. Census Bureau’s Business Trends and Out-
16 look Survey (BTOS) has asked a rotating, nationally representative sample
17 of roughly 200,000 employer businesses every two weeks whether they used
18 AI, producing the highest-frequency official adoption series ever published
19 for a general-purpose technology ([Bonney et al., 2024](#)). Firms maintain com-
20 parable internal telemetry on their own AI deployments that updates even
21 faster. Adoption evidence now arrives as a stream, and it is consumed as
22 one: dashboards are refreshed at every release, and decisions to scale, hold,
23 or stop technology investments are implicitly revisited at each refresh.

24 The inferential toolkit of technology and innovation management (TIM)
25 research was not built for this regime. The diffusion literature—from the
26 epidemic and probit traditions ([Bass, 1969](#); [Geroski, 2000](#); [Rogers, 2003](#)) to
27 contemporary firm-level studies of AI uptake ([McElheran et al., 2024](#); [Zolas
28 et al., 2020](#))—estimates adoption curves retrospectively and tests hypotheses
29 once, on a closed sample. When the same tests are applied repeatedly to an
30 accumulating series, as any monitoring practice implicitly does, their guar-
31 antees dissolve: the probability of at least one spurious “significant” finding
32 grows without bound in the number of looks, a fact established in sequential
33 analysis decades ago ([Wald, 1945](#); [Armitage et al., 1969](#)) and rediscovered
34 in industrial online experimentation as the “peeking problem” ([Johari et al.,
35 2022](#)). In this paper’s empirical setting the problem is not hypothetical: we
36 show that a standard rolling-window test applied to the BTOS AI series at

every release generates alarms on 37 of 48 possible looks, and on 31 of 48 even after Bonferroni correction, rendering “significance” useless as a signal of regime change.

A second obstacle is that high-frequency adoption measurement is itself unstable. On 17 November 2025 the Census Bureau replaced the BTOS question on AI use “in producing goods or services” with AI use in “any of its business functions.” The published national adoption rate moved from 10.0 percent in the last legacy-wording wave to 17.3 percent in the first revised-wording wave—a jump of 7.3 percentage points, or 25.7 design-based standard errors, that dwarfs every genuine movement in the series and reflects wording, not diffusion (Bick, 2026). Any credible monitoring scheme must separate shifts in the *latent diffusion state* from interventions on the *measurement apparatus*; the distinction is inherently causal (Pearl, 2009), and purely statistical change detectors cannot draw it.

We address both obstacles jointly and ask three research questions:

- **RQ1.** Can regime shifts in technology diffusion be detected in near-real time with error guarantees that remain valid under continuous, open-ended monitoring?
- **RQ2.** How can genuine shifts in the latent diffusion state be distinguished, in real time, from changes in the survey instrument that measures it?
- **RQ3.** What does such a monitor reveal about the dynamics of digital technology diffusion in the United States—with enterprise AI over 2023–2025 as the high-frequency focal case, set against the long view

61 since 1990—and how does its behavior compare with the repeated-
62 testing practices embedded in current monitoring conventions?

63 Our framework has two components. The first is a causal measurement
64 layer: the observed adoption rate is generated by a latent diffusion state
65 together with an explicit measurement-regime variable, and instrument re-
66 designs are modeled as interventions—in the do-operator sense—on the mea-
67 surement regime alone. Pre-announced redesign waves contribute no evidence
68 about diffusion, and a cross-strata diagnostic rooted in invariance logic (Pe-
69 ters et al., 2016; Rothenhäusler et al., 2021) verifies the attribution: an
70 instrument event moves all strata simultaneously in a single wave, whereas
71 genuine diffusion shifts build gradually and heterogeneously. The second
72 component is an e-process monitor. E-values and safe anytime-valid infer-
73 ence (Shafer and Vovk, 2019; Vovk and Wang, 2021; Howard et al., 2021;
74 Ramdas et al., 2023; Grünwald et al., 2024) provide nonnegative processes
75 that are martingales under the null hypothesis of “no departure from the
76 baseline adoption path,” so that by Ville’s inequality the probability that
77 the process *ever* crosses $1/\alpha$ under the null is at most α , regardless of how
78 often the analyst looks or when monitoring stops. We instantiate the moni-
79 tor with Robbins-type one-sided normal-mixture e-processes (Robbins, 1970)
80 on adoption increments standardized by the survey’s published design-based
81 standard errors, organized into surveillance epochs with re-anchoring after
82 each declared shift, in the spirit of sequential change detection (Page, 1954;
83 Shin et al., 2023).

84 Applying the framework to all 56 published BTOS waves through Decem-
85 ber 2025, and retrospectively to three decades of U.S. digital diffusion, yields

86 four substantive findings. First, the monitor declares a diffusion takeoff on
 87 20 April 2025 with terminal e-value 42.2, after an evidence trajectory that
 88 surged to 15.4 by 17 December 2023 and reached a local peak of 17.1 on
 89 28 January 2024, collapsed to 0.4 by 7 April 2024 during the first-half-2024
 90 growth stall (when measured adoption slipped from 4.9 to 4.7 percent), and
 91 rebuilt through the late-2024 and early-2025 acceleration. Second, against
 92 the anchored early-2025 growth rate of 0.299 percentage points per wave,
 93 the downward monitor quantifies the widely discussed mid-2025 decelera-
 94 tion at a peak evidence ratio of 24.3-to-1 on 7 September 2025—a strong
 95 “amber” that never crosses the declaration threshold; the restraint is vindi-
 96 cated within weeks as growth resumes to 10.0 percent, and it contrasts with
 97 a moving-average heuristic that whipsaws twice through the same episode.
 98 Third, the November 2025 redesign, the largest movement in the series by an
 99 order of magnitude, is excluded by design and confirmed as a measurement
 100 event by its signature: synchronized single-wave jumps of 6.1 to 13.8 stan-
 101 dard errors across all seven employment-size classes, against within-regime
 102 95th-percentile movements of 1.7 to 2.9. Fourth, a retrospective application
 103 of the same surveillance logic—with a predictable plug-in variance in place of
 104 design-based standard errors—recovers the punctuated evolutionary record
 105 of U.S. digital diffusion since 1990: on the annual internet-penetration se-
 106 ries it declares the takeoff in 1995 and the onset of saturation in 2008, and
 107 on the quarterly e-commerce share of retail sales it declares the takeoff in
 108 2004Q1, a mid-2010s acceleration in 2018Q1, the COVID-19 shock in the
 109 very quarter it occurred (2020Q2), and the post-pandemic re-acceleration in
 110 2023Q1, while flagging a single-year 2016 discontinuity in the internet se-

111 ries whose provenance cannot be settled without measurement metadata—a
112 long-view illustration of exactly why the causal measurement layer matters.
113 A companion timeline (Figure 7) aligns the intermediate technology waves
114 of the analytics era—data mining, cloud computing, big data, data science,
115 and deep learning—with the measured record, dated by canonical landmarks
116 because no official penetration series were ever produced for them.

117 The paper contributes to TIM methodology by importing anytime-valid
118 inference into diffusion research, replacing repeated testing with a procedure
119 whose error control survives the way monitoring is actually practiced; to
120 diffusion theory by formalizing survey redesigns as causal interventions on
121 the observation process, converting a measurement-comparability nuisance
122 (Crane et al., 2025) into an identified model component; and to practice by
123 distilling the machinery into a decision protocol—graded evidence thresholds
124 mapped to scale/hold/stop actions and re-anchoring rules—that applies as
125 readily to a firm’s internal adoption telemetry as to official statistics. In the
126 vocabulary of this special issue, the framework operationalizes an evolution-
127 ary view of industry dynamics: the punctuated uptake of a social technology
128 is tracked as it unfolds, as a sequence of dated, error-controlled regime dec-
129 larations rather than a curve fitted in hindsight.

130 Section 2 reviews the relevant literatures. Section 3 develops the method-
131 ology. Section 4 describes the data. Section 5 reports results, Section 6
132 discusses implications and limitations, and Section 7 concludes.

133 2. Theoretical background and related literature

134 2.1. *Measuring technology diffusion: from annual curves to biweekly streams*

135 Classical diffusion research models cumulative adoption as a smooth curve—
136 logistic or Bass-type under epidemic assumptions (Bass, 1969), or the aggre-
137 gate of heterogeneous threshold decisions in probit-style models (Geroski,
138 2000; Rogers, 2003)—while firm-level adoption studies in the technology-
139 acceptance tradition explain cross-sectional variation in who adopts (Davis,
140 1989). These literatures share a data regime: adoption observed annually
141 or less often, on closed samples, with inference conducted once after data
142 collection ends.

143 AI has broken that regime. The BTOS delivers a biweekly, nationally
144 representative firm-level adoption series (Bonney et al., 2024); the Annual
145 Business Survey provides deeper but slower benchmarks (Zolas et al., 2020;
146 McElheran et al., 2024); and household surveys track worker-side uptake of
147 generative AI at high frequency (Bick et al., 2025). Substantively, adoption
148 is fast and uneven: BTOS-based studies document use rates that more than
149 doubled within two years of the series’ launch, strong firm-size gradients,
150 and sectoral concentration in Information, Finance, and Professional Services
151 (U.S. Census Bureau, 2026a; Garcia Luna, 2026). Comparisons across instru-
152 ments reveal wide dispersion in point estimates—from about 5 to 40 percent
153 depending on question framing, respondent, and sampling unit (Crane et al.,
154 2025)—and the November 2025 BTOS redesign showed that wording alone
155 can nearly double a measured national adoption rate (Bick, 2026; Federal
156 Reserve Board, 2026). The literature has treated these measurement dis-
157 crepancies as a comparability nuisance to be footnoted; we treat them as

158 first-class objects in a causal model of the observation process.

159 2.2. *The peeking problem in adoption monitoring*

160 When an analyst inspects an accumulating series at every new wave and
161 applies a fixed-horizon test—rolling-window mean comparisons, regression
162 breaks, control-chart rules—each inspection is an implicit hypothesis test,
163 and the family-wise error rate compounds. [Armitage et al. \(1969\)](#) showed
164 that repeated significance testing at nominal level 0.05 produces a true false-
165 positive probability that grows without bound in the number of looks; [Johari](#)
166 [et al. \(2022\)](#) documented the same phenomenon in industrial A/B testing and
167 introduced always-valid p -values as a remedy. Group-sequential corrections
168 from clinical trials require the number and timing of looks to be fixed in
169 advance, an assumption that fits neither dashboards nor a survey program
170 whose supplements and redesigns arrive unpredictably. Technology-adoption
171 monitoring sits squarely in the uncorrected regime: to our knowledge, no
172 published study of AI diffusion adjusts its inferences for the sequential, open-
173 ended nature of the underlying data collection.

174 2.3. *E-values and anytime-valid inference*

175 E-values provide a measure of evidence against a null hypothesis with a
176 betting interpretation: an e-variable is a nonnegative random variable with
177 expectation at most one under the null, and an e-process is a nonnegative
178 process upper-bounded by a supermartingale with initial value one under
179 the null ([Shafer and Vovk, 2019](#); [Vovk and Wang, 2021](#); [Ramdas et al.,](#)
180 [2023](#); [Grünwald et al., 2024](#)). Ville’s inequality then yields time-uniform

181 error control: $\mathbb{P}_{H_0}(\sup_t E_t \geq 1/\alpha) \leq \alpha$. Evidence accumulates multiplica-
 182 tively, monitoring may stop or continue at any data-dependent time, and
 183 rejecting when $E_t \geq 1/\alpha$ controls the type-I error at α *uniformly over time*.
 184 Concrete constructions include Robbins’s method of mixtures for Gaussian
 185 observations (Robbins, 1970), time-uniform confidence sequences (Howard
 186 et al., 2021), and betting-based procedures for bounded random variables
 187 (Waudby-Smith and Ramdas, 2024); e-process analogues of CUSUM sup-
 188 port sequential changepoint detection with restarts (Page, 1954; Shin et al.,
 189 2023). These tools were developed for clinical monitoring and online exper-
 190 imentation; their application to macro-level technology diffusion appears to
 191 be new.

192 2.4. Causal structure of the measurement process

193 Structural causal models (Pearl, 2009) distinguish changes in a system’s
 194 mechanisms from changes in how the system is observed. Invariant causal
 195 prediction (Peters et al., 2016) and anchor regression (Rothenhäusler et al.,
 196 2021) operationalize a related idea: relationships that are structural re-
 197 main invariant across environments and interventions, whereas artifacts shift
 198 with the environment. In our setting the “environments” are measurement
 199 regimes—question wordings, panel compositions, supplement windows—and
 200 the invariance logic runs in a diagnostic direction: a genuine diffusion shift
 201 should be visible *within* a fixed measurement regime and build heteroge-
 202 neously *across* strata according to their exposure to the underlying driver,
 203 while a measurement intervention produces a discontinuity synchronized with
 204 the redesign date and common to all strata in a single wave. Section 3 em-
 205 beds this logic in the monitor itself, so that evidence accrued by the e-process

206 concerns the diffusion state, not the instrument.

207 3. Methodology

208 Figure 1 summarizes the framework. A biweekly adoption stream feeds a
209 causal measurement layer that excludes instrument-intervention waves and
210 standardizes within-regime increments; one-sided mixture e-processes accu-
211 mulate anytime-valid evidence against a baseline-path null; threshold cross-
212 ings feed a managerial decision protocol; and each declared shift triggers
213 re-anchoring of the monitor for a new surveillance epoch.

214 3.1. Observation model and null hypotheses

215 Let $t = 1, 2, \dots$ index biweekly survey waves within a measurement
216 regime. The published series delivers a weighted adoption-rate estimate $\hat{\pi}_t$
217 (percent of businesses reporting AI use in the past two weeks) with design-
218 based standard error $\hat{\sigma}_t$, together with parallel series for planned adoption
219 and for strata $k = 1, \dots, K$ (employment-size classes, sectors, states). We
220 posit a latent diffusion state θ_t —the population adoption propensity under a
221 fixed, ideal instrument—and a measurement regime M_t recording the active
222 question wording and panel design. The observation equation is

$$\hat{\pi}_t = g(\theta_t, M_t) + \varepsilon_t, \quad \varepsilon_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, \hat{\sigma}_t^2), \quad (1)$$

223 where g maps the latent state through the instrument (a narrower wording
224 truncates θ_t to its production-function component) and the Gaussian ap-
225 proximation for the sampling error ε_t is justified by the survey’s scale: each
226 biweekly estimate aggregates roughly 15,000–20,000 weighted responses ([Fed-
227 eral Reserve Board, 2026](#)), and consecutive waves are drawn from different

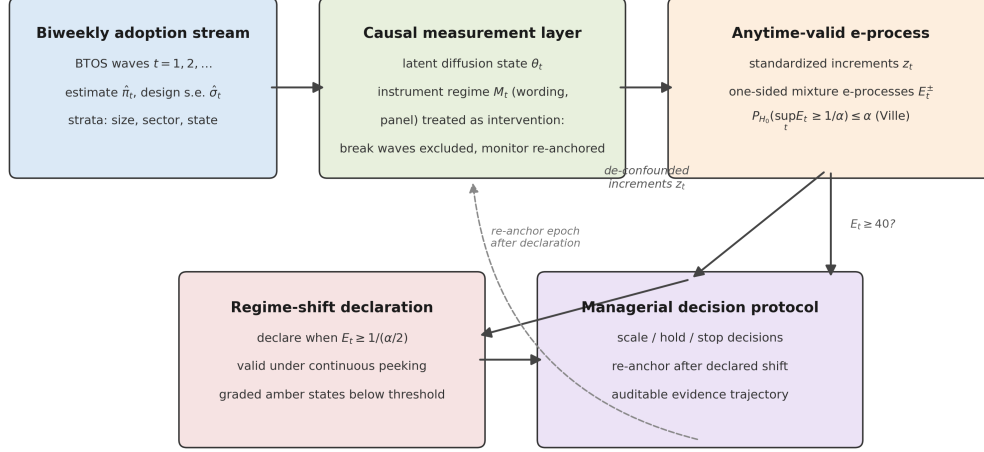


Figure 1: The causal e-process monitoring framework. The measurement layer separates the latent diffusion state θ_t from the instrument regime M_t ; one-sided mixture e-processes accumulate time-uniformly valid evidence on standardized within-regime increments; declared shifts and instrument changes trigger re-anchoring.

228 rotation panels, so sampling errors are independent across waves. Instru-
 229 ment redesigns are interventions $\text{do}(M_t = m')$ on the observation process:
 230 they change g , not θ_t . The monitoring question is whether θ_t has departed
 231 from a baseline path—not whether $\hat{\pi}_t$ has jumped.

232 Within a regime, define the increment $\Delta_t = \hat{\pi}_t - \hat{\pi}_{t-1}$. A surveillance
 233 epoch e is endowed with a *baseline slope* β_e (percentage points per wave),
 234 fixed at the epoch’s start from information available at that time. The
 235 epoch’s null hypothesis is that the latent path continues at the baseline rate:

$$H_0^{(e)} : \quad \mathbb{E}[\Delta_t \mid \mathcal{F}_{t-1}] = \beta_e \quad \text{with} \quad \Delta_t - \beta_e \sim \mathcal{N}(0, \sigma_t^2), \quad \sigma_t^2 = \hat{\sigma}_t^2 + \hat{\sigma}_{t-1}^2 + \text{se}(\hat{\beta}_e)^2, \quad (2)$$

236 where $\text{se}(\hat{\beta}_e)$ accounts for estimation error in an anchored slope (zero when

237 β_e is pre-registered rather than estimated). Under $H_0^{(e)}$ the standardized
 238 increments $z_t = (\Delta_t - \beta_e)/\sigma_t$ are independent standard normal. Rejecting
 239 $H_0^{(e)}$ upward is evidence of takeoff or acceleration relative to the baseline;
 240 rejecting downward is evidence of deceleration or stagnation. The null is
 241 deliberately sharp: it attributes all within-regime dispersion to sampling
 242 error, so that *any* persistent movement of the latent path away from the
 243 baseline counts as signal. Section 5.8 examines robustness to relaxing this
 244 attribution.

245 3.2. One-sided mixture e -processes

246 For each epoch we run two one-sided monitors built from Robbins's
 247 method of mixtures (Robbins, 1970). Let $S_n = \sum_{s=1}^n z_s$ denote the cumu-
 248 lative standardized evidence after n within-epoch increments and let $\tau > 0$
 249 be a prior scale. Mixing the Gaussian likelihood ratio $\exp(\delta S_n - n\delta^2/2)$ over
 250 a half-normal prior on the drift $\delta > 0$ gives, in closed form (derivation in
 251 Appendix A),

$$E_n^+ = \frac{2}{\tau\sqrt{a_n}} \Phi\left(\frac{S_n}{\sqrt{a_n}}\right) \exp\left(\frac{S_n^2}{2a_n}\right), \quad a_n = n + \tau^{-2}, \quad (3)$$

252 with $E_0^+ = 1$, and symmetrically E_n^- obtained by replacing S_n with $-S_n$.
 253 Under $H_0^{(e)}$, (E_n^+) and (E_n^-) are nonnegative martingales with initial value
 254 one; more generally, under the composite one-sided null $\mathbb{E}[z_t \mid \mathcal{F}_{t-1}] \leq 0$
 255 (with unit-variance Gaussian noise), (E_n^+) is a supermartingale (Appendix
 256 A). Ville's inequality therefore gives the time-uniform guarantee

$$\mathbb{P}_{H_0}\left(\exists n : E_n^+ \geq \frac{1}{\alpha/2}\right) \leq \frac{\alpha}{2}, \quad (4)$$

and likewise for E_n^- , so that declaring a regime shift the first time *either* monitor reaches $1/(\alpha/2) = 40$ (at $\alpha = 0.05$) controls the per-epoch false-declaration probability at α no matter how often the series is inspected or when monitoring stops. Throughout we set $\tau = 1$, a unit prior scale on the standardized drift; Section 5.8 shows the results are insensitive to this choice. Because E_n is a likelihood-ratio-type quantity, intermediate values carry graded meaning— $E_n = 10$ is 10-to-1 evidence against the baseline—which motivates the “amber” states of the decision protocol below, an interpretive resource that repeated p -values cannot supply under continuous monitoring.

3.3. Surveillance epochs and re-anchoring

Monitoring proceeds through epochs, in the restart spirit of CUSUM (Page, 1954) and its e-process descendants (Shin et al., 2023):

1. **Epoch 1 (takeoff surveillance).** The baseline slope is pre-registered at $\beta_1 = 0$: the null is that the latent adoption path is flat. Both one-sided monitors run from the first increment.
2. **Declaration.** The first time $E_n^+ \geq 40$ or $E_n^- \geq 40$, a regime shift is declared with its direction, date, and terminal e-value recorded.
3. **Re-anchoring.** The next epoch’s baseline slope $\hat{\beta}_{e+1}$ and its standard error are estimated by weighted least squares (weights $1/\hat{\sigma}_t^2$) on the $w = 8$ waves ending at the declaration wave—a window that uses only past data, so predictability and hence the supermartingale property are preserved. The new epoch begins at the following wave with fresh monitors at wealth one.

281 **4. Instrument interventions.** At a pre-announced redesign wave ($M_t \neq$
282 M_{t-1}), the transition increment is excluded from every monitor—the
283 do-operator made operational—and the monitor re-anchors on post-
284 redesign waves once w of them have accumulated. For discontinuities
285 that are *not* pre-announced, a cross-strata diagnostic (Section 3.4) al-
286 locates the wave to the measurement layer or the diffusion layer before
287 any evidence is booked.

288 3.4. Cross-strata attribution diagnostic

289 Let $z_{k,t}$ be the standardized increment in stratum k at wave t . The di-
290 agnostic compares the profile of a candidate break wave against the stratum’s
291 own within-regime history: an instrument event exhibits (i) synchronization—
292 every stratum jumps in the same wave; (ii) extremity—each stratum’s $|z_{k,t}|$
293 far exceeds the upper tail (we report the 95th percentile) of its within-regime
294 $|z|$ distribution; and (iii) sign coherence consistent with the mechanical ef-
295 fect of the instrument change (a broader wording can only raise measured
296 adoption). A genuine diffusion shift, by contrast, builds over multiple waves
297 with magnitudes inside or near the within-regime tail and loads heteroge-
298 neously on strata according to exposure. The diagnostic is a screening rule
299 with a causal-invariance rationale (Peters et al., 2016; Rothenhäusler et al.,
300 2021) rather than a formally identified estimator; in the application it is
301 unambiguous (Section 5.3).

302 3.5. Retrospective surveillance with plug-in variance

303 The BTOS application enjoys a luxury most diffusion series lack: pub-
304 lished design-based standard errors that make the sampling distribution

305 of increments known. To extend surveillance to historical series without
 306 such metadata (Section 4.2), we replace σ_t in (2) with a *predictable plug-in*
 307 scale: $\hat{\sigma}_t = \max\{\phi, 1.4826 \cdot \text{MAD}(r_{t-W}, \dots, r_{t-1})\}$, the median-absolute-
 308 deviation estimate over the trailing $W = 8$ within-epoch increment residuals
 309 $r_s = \Delta_s - \beta_e$, seeded at each re-anchoring from the anchoring window’s own
 310 residuals and floored at a series-specific constant ϕ chosen conservatively
 311 (0.8 percentage points per year for the annual internet series; 0.08 points per
 312 quarter for the quarterly e-commerce series). Because the scale at wave t uses
 313 only information through $t - 1$, predictability—and hence the supermartin-
 314 gale structure—is preserved mechanically; but the Gaussian null in (2) now
 315 carries an estimated rather than known variance, so the time-uniform guaran-
 316 tee (4) holds only approximately. We therefore treat the retrospective results
 317 as an illustrative application of the surveillance logic, report e-values above
 318 10^4 simply as decisive, and reserve exact error statements for the design-based
 319 BTOS case.

320 3.6. Decision protocol

321 Table 1 maps monitor states to actions, separating *statistical* thresholds
 322 (fixed by α and the e-process) from *managerial* ones (encoding the asymme-
 323 try of scaling too early versus too late, and settable per organization).

324 3.7. Comparators

325 Three baselines represent current monitoring practice, each applied at
 326 every release to the naively spliced series (legacy waves followed by revised-
 327 wording waves, as a dashboard unaware of the redesign would splice them):
 328 (a) a repeated two-sample z -test comparing the latest four waves with the

Table 1: Decision protocol linking e-process states to managerial actions ($\alpha = 0.05$; declaration threshold $1/(\alpha/2) = 40$ per one-sided monitor).

Monitor state	Evidence level	Interpretation	Illustrative action
$E_n < 2$	Negligible	No credible departure from baseline path	Continue routine monitoring
$2 \leq E_n < 10$	Amber	Suggestive evidence; odds 2–10:1 against baseline	Prepare contingency; increase review cadence
$10 \leq E_n < 40$	Strong amber	Odds 10–40:1; declaration plausible soon	Pre-position decision; brief stakeholders
$E_n \geq 40$	Declared shift	Time-uniformly valid rejection at level α	Execute scale/hold/stop decision; re-anchor monitor
Instrument change ($M_t \neq M_{t-1}$)	—	Measurement intervention, not diffusion evidence	Exclude wave; re-baseline; document break

329 preceding four, using the design-based standard errors, uncorrected at the
 330 5% level; (b) the same with a Bonferroni correction over the number of looks
 331 made so far; and (c) a moving-average crossing rule (3-wave minus 8-wave
 332 means changing sign), the de facto dashboard heuristic. For each we record
 333 the first alarm, the number of alarm waves and of distinct alarm episodes
 334 over the 48 possible looks, and behavior at the November 2025 redesign.

335 4. Data

336 4.1. The Business Trends and Outlook Survey, 2023–2025

337 The BTOS is an experimental data product of the U.S. Census Bureau
338 designed to measure business conditions at high frequency. The sample
339 comprises approximately 1.2 million nonfarm employer businesses divided
340 into six rotating panels of roughly 200,000; each panel reports once every
341 twelve weeks over a year, and collection occurs every two weeks, yielding
342 a continuous biweekly national series ([U.S. Census Bureau, 2025, 2026b](#)).
343 Estimates are weighted to be representative at the national, state, sector,
344 and employment-size levels; the average biweekly response rate over the AI-
345 content period is about 16 percent ([Bonney et al., 2024](#)), and estimates failing
346 publication standards are suppressed.

347 Two core AI questions have run since the collection period beginning 11
348 September 2023 ([Bonney et al., 2024](#)). From September 2023 through Oc-
349 tober 2025 the current-use question read: *“In the last two weeks, did this*
350 *business use Artificial Intelligence (AI) in producing goods or services? (Ex-*
351 *amples of AI: machine learning, natural language processing, virtual agents,*
352 *voice recognition, etc.)”* Beginning with the collection of 17 November 2025,
353 “in producing goods or services” was replaced by “in any of its business
354 functions,” and a supplement fielded from 17 November 2025 to 8 February
355 2026 added detail on adoption across fifteen business functions ([U.S. Census](#)
356 [Bureau, 2026b](#)). A companion question asks about expected AI use over the
357 next six months under the same wordings.

358 Our analysis dataset consists of every published national and employment-
359 size-class estimate of the two AI questions, with design-based standard errors,

360 from the survey’s public data downloads. Because the Census Bureau’s file
361 of 15 December 2025 removed the legacy-wording AI rows when introducing
362 the revised series, we splice two archived vintages of the official National
363 and Employment Size Class files—downloaded 8 December 2025 (which re-
364 tains the full legacy series) and 15 December 2025 (which carries the revised
365 series)—following the vintage documentation of [Goldschlag \(2025\)](#). The re-
366 sult is 56 national waves: 54 legacy-wording waves covering collection periods
367 ending 24 September 2023 through 5 October 2025 (periods 202319–202520),
368 and two revised-wording waves ending 30 November and 14 December 2025
369 (202524–202525). No AI estimates were published for the three transition
370 periods 202521–202523. National design-based standard errors have median
371 0.15 percentage points (range 0.08–0.23); size-class standard errors are larger,
372 with medians from 0.23 (1–4 employees) to 1.21 (250+ employees) percent-
373 age points. The exact analysis dataset (two CSV files) and replication code
374 accompany the manuscript.

375 The headline contours of the series are as follows. Under the legacy
376 wording, national current use rose from 3.7 percent (24 September 2023) to
377 4.9 percent by end-2023, slipped to 4.7 percent by 30 June 2024, resumed
378 growth to 6.3 percent by 29 December 2024, accelerated to 8.3 percent by 20
379 April 2025, flattened near 8.8 percent through 10 August 2025, and closed
380 the legacy period at 10.0 percent on 5 October 2025. Planned use ran above
381 current use throughout, from 6.3 to 14.0 percent. Under the revised wording,
382 current use printed 17.3 and 17.2 percent and planned use 21.1 and 20.9
383 percent in the two available waves—consistent with subsequently published
384 Census reporting that the broadened measure hovered between 17 and 20

385 percent into 2026 ([U.S. Census Bureau, 2026a](#)).

386 4.2. Long-run diffusion series, 1990–2026

387 Two public series extend the surveillance horizon to three and a half
388 decades. *Internet users as a share of the U.S. population*, annual, 1990–2024,
389 from the International Telecommunication Union’s World Telecommunica-
390 tion/ICT Indicators database as disseminated by the World Bank (indicator
391 IT.NET.USER.ZS) and retrieved from FRED ([World Bank, 2026](#)): the series
392 rises from 0.79 percent in 1990 to 94.7 percent in 2024, tracing the canonical
393 S-curve of a saturating general-purpose technology. It splices household-
394 survey estimates with ITU and World Bank estimation methods, so—unlike
395 the BTOS—it carries neither design-based standard errors nor strata, and
396 it contains a conspicuous single-year discontinuity of +10.9 points between
397 2015 and 2016 following a decade in which measured penetration fluctuated
398 between 69 and 75 percent. *E-commerce sales as a share of total U.S. retail*
399 *sales*, quarterly and seasonally adjusted, 1999Q4–2026Q1, from the Census
400 Bureau’s Quarterly Retail E-Commerce Sales program retrieved from FRED
401 ([U.S. Census Bureau, 2026c](#)): 106 quarters from 0.6 to 16.9 percent, including
402 the celebrated COVID-19 jump (11.9 to 16.3 percent between 2020Q1 and
403 2020Q2), the 2021–2022 partial reversal to 14.2 percent, and renewed growth
404 thereafter. Both extracts are included verbatim in the supplementary data.

405 5. Results

406 5.1. Episode 1: the takeoff and its declaration (RQ1, RQ3)

407 Figures 2 and 3 display the series and the monitors’ evidence trajectories.
408 Epoch 1 opens at the second published wave (8 October 2023) against the pre-

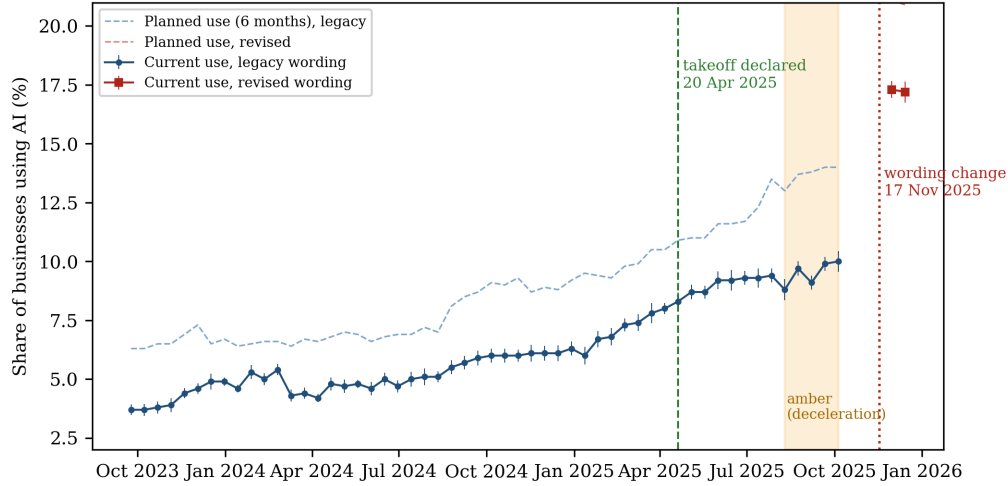


Figure 2: The BTOS national AI adoption series, September 2023–December 2025, with monitoring outcomes. Solid lines with 95% intervals show current use under the legacy (blue circles) and revised (red squares) wordings; dashed lines show planned use over the next six months. The green dashed line marks the takeoff declaration (20 April 2025); the shaded band marks the span during which the deceleration monitor held amber-or-higher evidence (10 August–5 October 2025); the red dotted line marks the question-wording intervention (17 November 2025). No AI estimates were published for the three transition periods between 5 October and 30 November 2025. Source: authors’ calculations from U.S. Census Bureau BTOS public data.

409 registered flat baseline. Evidence accumulates quickly through the late-2023
 410 growth spurt: the upward monitor passes the amber threshold ($E \geq 2$) on
 411 19 November 2023, passes 10 on 17 December 2023 ($E = 15.4$), and reaches
 412 a local peak of 17.1 on 28 January 2024. It then does something a p -value
 413 stream cannot do: it retreats. As measured adoption stalls through the first
 414 half of 2024 (4.9 percent at end-2023 to 4.7 percent on 30 June 2024), the
 415 negative increments consume accumulated wealth, and the monitor bottoms

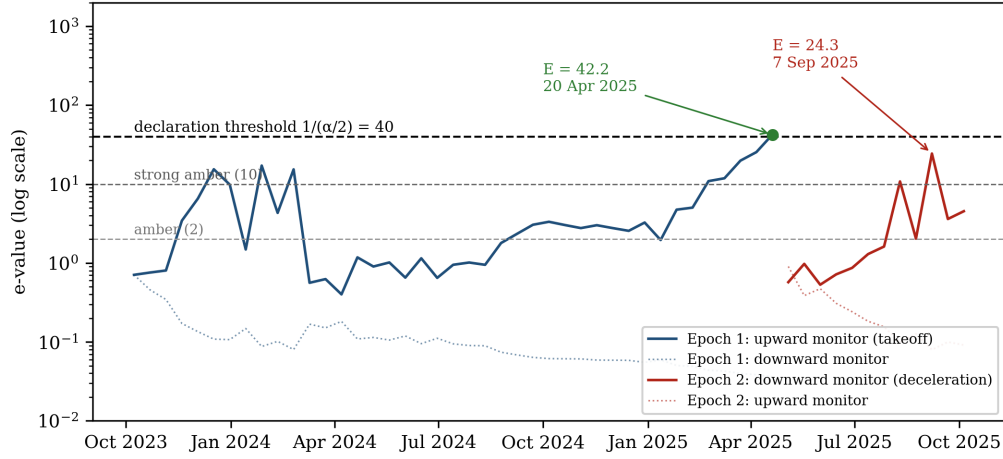


Figure 3: E-value trajectories of the national monitors (log scale). Epoch 1 tests the pre-registered flat baseline ($\beta_1 = 0$); Epoch 2 tests the anchored early-2025 growth rate ($\hat{\beta}_2 = 0.299 \pm 0.027$ pp per wave, estimated on the eight waves ending at the declaration). The upward monitor crosses the declaration threshold of 40 on 20 April 2025 ($E = 42.2$); the Epoch-2 downward monitor peaks at $E = 24.3$ on 7 September 2025 without crossing.

416 at $E = 0.4$ on 7 April 2024—the evidence for a takeoff, evaluated against
 417 the full history, had genuinely weakened. The monitor rebuilds through
 418 the second-half-2024 recovery (6.3 percent by 29 December 2024) and the
 419 early-2025 acceleration (6.0 percent on 12 January to 8.3 percent on 20 April
 420 2025), and crosses the declaration threshold on **20 April 2025** with terminal
 421 e-value **42.2**, after 41 monitored increments. By Ville’s inequality (4), the
 422 probability that a flat latent path would ever have produced such a crossing
 423 is at most 2.5 percent, no matter how many times the series was inspected
 424 along the way.

425 The declaration date is informative rather than embarrassing. The mon-
 426 itor did not certify a takeoff during the 2023 spurt because, at that point, a

427 flat path punctuated by a transient could not yet be excluded at the stated
 428 error rate—and the first half of 2024 showed exactly such a stall. The decla-
 429 ration arrives precisely when the acceleration becomes sustained. The answer
 430 to RQ1 is therefore affirmative in a specific sense: real-time detection with
 431 time-uniform guarantees is feasible on official high-frequency adoption data,
 432 and the guarantee’s price is measured restraint during ambiguous phases, not
 433 blindness.

434 5.2. Episode 2: the mid-2025 deceleration as graded evidence (RQ3)

435 Upon declaration, the monitor re-anchors: weighted least squares on the
 436 eight waves ending at the declaration (12 January–20 April 2025) fixes the
 437 Epoch-2 baseline at $\hat{\beta}_2 = 0.299 \pm 0.027$ percentage points per wave—the
 438 null that the early-2025 growth tempo continues. Epoch 2 opens on 4 May
 439 2025. Against this demanding baseline, the flattening of the series through
 440 the summer (8.3 percent in April to 8.8 percent on 10 August 2025) drives
 441 the *downward* monitor upward: it reaches amber on 10 August 2025, and
 442 peaks at $E = 24.3$ on 7 September 2025—24-to-1 evidence that growth had
 443 fallen below the anchored tempo, within the strong-amber band of Table 1
 444 but short of declaration. The evidence then recedes to $E = 4.5$ by 5 October
 445 2025 as growth resumes (9.9 and 10.0 percent in the final legacy waves). The
 446 upward monitor never exceeds 0.9 in Epoch 2.

447 Two readings of this episode matter for practice. First, the monitor
 448 *quantified* the widely discussed 2025 deceleration in real time—at its peak, a
 449 manager consulting the dashboard would have seen 24-to-1 odds that AI dif-
 450 fusion momentum had broken—while correctly declining to certify a regime
 451 change that the subsequent data contradict: the series re-accelerated within

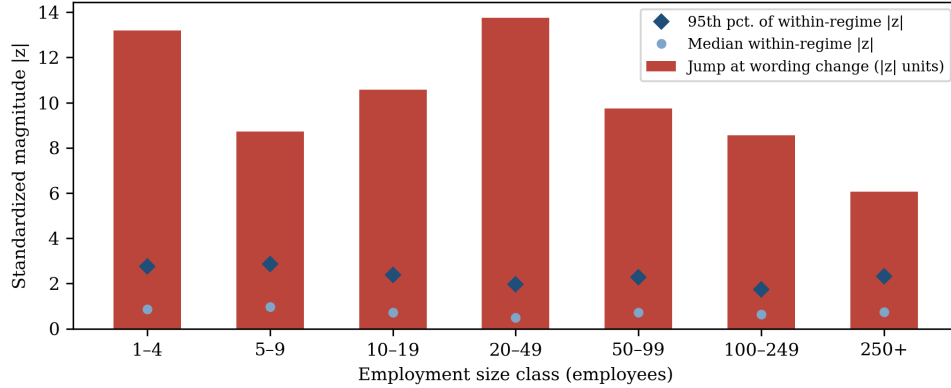


Figure 4: Cross-strata attribution diagnostic at the 17 November 2025 wording change. Bars show each employment-size class’s jump between its last legacy-wording and first revised-wording estimate, in units of the combined design-based standard error; markers show the median and 95th percentile of that class’s within-regime standardized wave-to-wave movements. Every class jumps simultaneously, by 2–7 times its within-regime 95th percentile, with a common sign—the signature of an instrument intervention, not of diffusion.

452 weeks, and the revised-wording waves and later Census reporting show con-
 453 tinued growth (U.S. Census Bureau, 2026a). Had the organization operated
 454 at $\alpha = 0.10$ instead of 0.05 (declaration threshold 20), a deceleration *would*
 455 have been declared on 7 September 2025 and then contradicted—a concrete
 456 illustration of what the choice of α buys. Second, the moving-average heuris-
 457 tic run over the same episode crossed downward on 7 September and back
 458 upward on 21 September 2025 (Figure 5): a whipsaw delivering two contra-
 459 dictory signals fourteen days apart, with no statement of error probability
 460 attached to either.

461 *5.3. Episode 3: the November 2025 redesign as a measurement event (RQ2)*

462 Between the last legacy wave (10.0 percent, 5 October 2025) and the
463 first revised-wording wave (17.3 percent, 30 November 2025) the published
464 series jumps 7.3 percentage points—25.7 combined standard errors, against
465 a within-regime national history in which the median absolute standard-
466 ized movement is 0.94 and the maximum is 6.2. Because the redesign was
467 announced by the data provider, the causal layer excludes the transition in-
468 crement by construction, and the monitor books no diffusion evidence from
469 it. The cross-strata diagnostic of Section 3.4, applied as if the break had
470 *not* been announced, confirms the attribution decisively (Figure 4 and Ta-
471 ble 3): all seven employment-size classes jump in the same single wave, by 6.1
472 to 13.8 standard errors—between roughly 2.6 and 7.0 times their respective
473 within-regime 95th percentiles (1.74–2.85)—and all in the same direction, as
474 the mechanically broader wording implies. Nothing in the genuine takeoff
475 resembles this signature: the takeoff built over more than a year, wave by
476 wave, with movements inside or near the within-regime tail.

477 The contrast with conventional practice is stark. The naive rolling z -test,
478 applied to the spliced series as a redesign-unaware dashboard would splice it,
479 produces its largest statistic of the entire sample at the break ($|z| = 32.9$) and
480 alarms there under both the uncorrected and Bonferroni-corrected versions—
481 certifying, at any conventional significance level, the largest “diffusion shock”
482 on record where none occurred. The answer to RQ2 is that the separation
483 is achievable and cheap: it requires only that the monitoring architecture
484 carry the instrument regime as an explicit variable and treat its changes as
485 interventions, with the cross-strata signature available as verification when

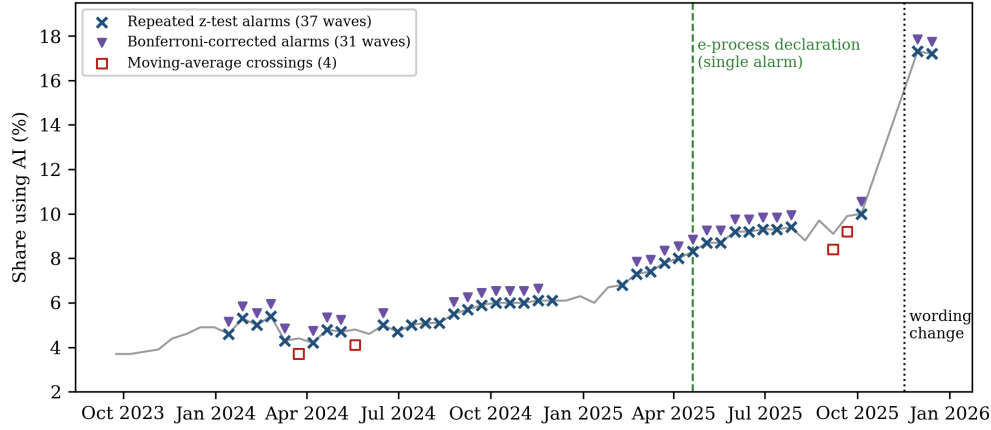


Figure 5: Alarm behavior of conventional monitoring rules over the naively spliced series, September 2023–December 2025. Crosses: repeated four-versus-four-wave z -test, uncorrected (37 alarm waves of 48 looks). Triangles: the same with Bonferroni correction over looks (31 alarm waves). Open squares: 3-versus-8-wave moving-average crossings (four crossings, including the September 2025 whipsaw). The e-process framework produces a single declaration (green dashed line) and excludes the wording change (dotted line) by design.

486 provenance is uncertain.

487 5.4. Comparison with conventional monitoring (RQ3)

488 Table 2 and Figure 5 summarize the comparison, and the honest read-
 489 ing cuts both ways. The repeated z -tests “detect” growth on 14 January
 490 2024, fifteen months before the e-process declaration—but the same rule
 491 then alarms on 37 of 48 looks (31 after Bonferroni), including through the
 492 first-half-2024 stall and at the November 2025 redesign. A rule that is almost
 493 always alarming cannot date a regime change, cannot distinguish diffusion
 494 from measurement, and offers no interpretable error statement under open-
 495 ended monitoring: the uncorrected version’s family-wise guarantee is void,

Table 2: Detection and alarm behavior across monitoring methods on the BTOS national AI series (48 possible looks, September 2023–December 2025).

Method	First alarm	Alarm waves	Distinct episodes	Alarms at redesign
Causal e-process ($\alpha = 0.05$)	20 Apr 2025 ^a	1	1	0 (excluded by design)
Repeated z -test, uncorrected	14 Jan 2024	37	5	yes ($ z = 32.9$)
Repeated z -test, Bonferroni	14 Jan 2024	31	6	yes
Moving-average crossing (3, 8)	24 Mar 2024	4	3	no crossing

^aDeclaration with terminal e-value 42.2; graded evidence available continuously beforehand (amber from 19 November 2023, strong amber from 17 December 2023, trough 0.4 on 7 April 2024). Moving-average crossing dates: 24 March 2024, 19 May 2024, 7 September 2025, 21 September 2025.

496 and the Bonferroni version’s guarantee, while formally valid look-by-look, is
497 purchased by a level that shrinks toward zero as monitoring continues, with
498 its 31 alarms demonstrating that in this high-precision setting the correc-
499 tion restrains nothing that matters. The moving-average rule is quiet by
500 comparison but delivers whipsaws (two crossings in 2024 during the stall-
501 recovery, two in September 2025) with no error control at all. The e-process
502 trades earliness for validity and interpretability: one declaration, dated, with
503 a terminal evidence ratio; a continuous graded evidence path before it; cor-
504 rect restraint during the plateau; and structural immunity to the instrument
505 change. Which side of that trade a decision-maker should take depends on
506 the cost of false regime declarations—and Section 6 argues that for scaling
507 decisions in technology management, those costs are high.

Table 3: Employment-size-class series and monitoring outcomes. “Max E^+ ” is the maximum of the takeoff (Epoch-1) upward monitor over the legacy period; no class reaches the declaration threshold of 40, reflecting larger stratum-level standard errors (median s.e. in percentage points shown). Jump columns give the movement between the last legacy-wording and first revised-wording estimates in percentage points and in combined standard errors, against the 95th percentile of within-regime standardized movements.

Size class	Sep 2023	Oct 2025	Nov 2025 ^r	Med. s.e.	Max E^+	Jump (pp)	Jump ($ z $)	p95 wit
1–4	4.3	11.0	17.0	0.23	11.4	6.0	13.2	2.7
5–9	3.0	8.6	15.5	0.29	4.0	6.9	8.7	2.8
10–19	2.6	8.1	17.8	0.32	10.7	9.7	10.6	2.3
20–49	2.6	7.0	18.7	0.36	1.8	11.7	13.8	1.9
50–99	2.4	9.7	20.7	0.57	4.0	11.0	9.7	2.2
100–249	2.5	8.8	23.7	0.92	1.3	14.9	8.6	1.7
250+	3.1	13.0	30.5	1.21	12.9	17.5	6.1	2.3

Entries in percent of businesses except where noted. ^rFirst revised-wording estimate (30 November 2025). Source: authors’ calculations from U.S. Census Bureau BTOS public data (Employment Size Class files, vintages of 8 and 15 December 2025).

5.5. Heterogeneity across firm-size classes

Table 3 reports the size-class analysis. Levels replicate the familiar gradient—by the revised measure, adoption runs from 17.0 percent among the smallest firms to 30.5 percent among firms with 250 or more employees, and the revised-wording jump is largest in absolute terms for large firms (17.5 percentage points), whose multi-function operations the legacy wording most severely under-counted. Stratum-level monitors, facing standard errors four to eight times the national ones, accumulate takeoff evidence that peaks in amber territory (maximum E^+ of 11.4–12.9 for the smallest and largest

517 classes, weaker in between) without any stratum-level declaration in-sample—
518 a transparent illustration that the framework’s detection power is governed
519 by the precision of the stream it watches, and that national-level surveillance,
520 which pools that precision, is where declarations are earned. For the attribu-
521 tion diagnostic, the strata are decisive in the other direction, as Section 5.3
522 showed: it is exactly because every class jumps at once, far outside its own
523 history, that the redesign is unmistakable as a measurement event.

524 5.6. *The long view: three diffusion waves, 1990–2026*

525 Figure 6 and Table 4 place the AI episode in its evolutionary context. Run
526 with the plug-in-variance monitor of Section 3.5, the surveillance recovers the
527 punctuated record of U.S. digital diffusion as a sequence of dated declara-
528 tions. For the internet, the takeoff is declared in 1995—the year commercial
529 access became mainstream—against the pre-registered flat baseline; against
530 the anchored boom-era growth rate of 6.75 points per year (1996–2003), the
531 downward monitor declares the onset of saturation in 2008; and no further
532 shift is declared through 2024 against the ensuing plateau-era baseline. For
533 e-commerce the sequence is denser: takeoff declared in 2004Q1; a genuine
534 acceleration in the growth *rate*—from about 0.10 to about 0.32 points per
535 quarter—declared in 2018Q1; the COVID-19 lockdown shock declared in the
536 very quarter it occurred, 2020Q2, with decisive evidence; and, against the
537 post-pandemic baseline of *declining* share (−0.11 per quarter, anchored on
538 2020Q3–2022Q2), the resumption of growth declared in 2023Q1. The moni-
539 tor is quiet where the series is quiet: no declaration stands open at the end
540 of either sample.

541 Two lessons carry back to the paper’s core argument. First, the COVID-

542 19 episode furnishes the natural counterpart to the November 2025 BTOS
 543 redesign: both are large single-period jumps, but the 2020Q2 shock is a
 544 genuine diffusion event—externally corroborated, persistent in level, and oc-
 545 ccurring within an unchanged measurement instrument—whereas the 2025
 546 jump is pure instrument. A monitoring architecture must be able to declare
 547 the first and absorb the second, and this one does. Second, the internet series
 548 contains the cautionary intermediate case: a +10.9-point jump in a single
 549 year (2015–2016) after a decade-long statistical plateau, a profile that under
 550 the diagnostic of Section 3.4 is highly suspect as a measurement artifact—the
 551 series splices household-survey estimates with ITU and World Bank estima-
 552 tion methods (World Bank, 2026)—but that cannot be adjudicated, because
 553 no design-based errors, strata, or documented redesign dates accompany it.
 554 In our epoch structure the jump falls inside the 2009–2016 anchoring window
 555 and therefore contaminates the plateau-era baseline rather than triggering a
 556 declaration; an analyst without the measurement layer would have been left
 557 to guess. The comparison across the three panels of Figure 6 is thus also a
 558 comparison across measurement regimes—annual splices, quarterly designed
 559 surveys, biweekly designed surveys with published errors and strata—and it
 560 makes concrete how much inferential reach the improvement in measurement
 561 infrastructure has purchased.

562 5.7. *The intermediate waves: data mining, cloud, big data, data science, and* 563 *deep learning*

564 Between the internet wave and the AI wave lie the technologies through
 565 which the analytics era actually arrived—data mining in the 1990s, cloud
 566 computing from 2006, big data and data science from the early 2010s, and

567 the deep-learning breakout of 2012–2015. Figure 7 puts all of them in one
568 picture, and it does so under an honesty constraint that is itself a find-
569 ing: *none* of these waves was ever tracked by an official penetration series.
570 No statistical agency asked businesses, at any frequency, whether they used
571 data mining in 1997, cloud infrastructure in 2009, or deep learning in 2016;
572 the first designed, recurring, error-quantified adoption instrument for any of
573 these general-purpose capabilities is the 2023 BTOS AI question set analyzed
574 in this paper. The intermediate waves are therefore dated in panel (a) by
575 canonical landmarks—the founding KDD conference and framework (Fayyad
576 et al., 1996), the EC2 launch and the Berkeley cloud retrospective (Armbrust
577 et al., 2010), the big-data research agenda (Chen et al., 2012), the codification
578 of data science (Provost and Fawcett, 2013), AlexNet and the deep-learning
579 consolidation (Krizhevsky et al., 2017; LeCun et al., 2015), and the Chat-
580 GPT release—rather than by measured curves, and we deliberately decline to
581 substitute proxies such as search-interest or bibliometric indices for adoption
582 shares within the monitoring framework, since their measurement processes
583 (query-share renormalization, corpus growth, indexing-policy changes) are
584 exactly the kind of undocumented instrument drift the framework exists to
585 quarantine.

586 Read jointly, the two panels tell an evolutionary story with a measurement
587 moral. The discourse waves cluster in the long gap between the internet’s
588 declared saturation onset (2008) and the AI takeoff (2025)—the period in
589 which the capability stack (storage, elastic compute, learning algorithms) was
590 assembled that the 2023–2025 AI diffusion then drew upon; the e-commerce
591 acceleration declared in 2018Q1 is the one place where an intermediate wave

leaves a footprint in a measured series, consistent with the maturation of cloud-era retail infrastructure in the mid-2010s. That the AI wave is the first to be observable biweekly, with design-based errors and strata, is why the monitoring guarantees of this paper became possible at all—and why, prospectively, the framework can be in place from a technology wave’s first measured wave rather than reconstructed after the fact.

5.8. *Robustness*

Two sensitivity checks probe the construction. First, the prior scale: re-running the full surveillance with $\tau = 0.5$ and $\tau = 2$ leaves the takeoff declaration on 20 April 2025 for $\tau \in \{0.5, 1\}$ (terminal e-values 58.8 and 42.2) and moves it one wave later, to 4 May 2025, for $\tau = 2$ ($E = 45.8$); the qualitative Epoch-2 pattern is unchanged. Second, the sharp attribution of within-regime dispersion to sampling error: inflating every σ_t by a factor $\kappa = 1.5$ —equivalent to positing that half again as much wave-to-wave noise is non-informational—prevents any declaration in-sample (the takeoff monitor peaks at 5.8). The framework’s conclusions are thus robust to the prior but, as they should be, sensitive to the variance model: the design-based standard errors are the published, defensible quantification of sampling noise for this survey, and analysts who believe substantial idiosyncratic latent wobble exists can encode that belief in κ and accept correspondingly later detection. We regard the transparency of that dial as a feature: it makes explicit an assumption that rolling t -tests bury.

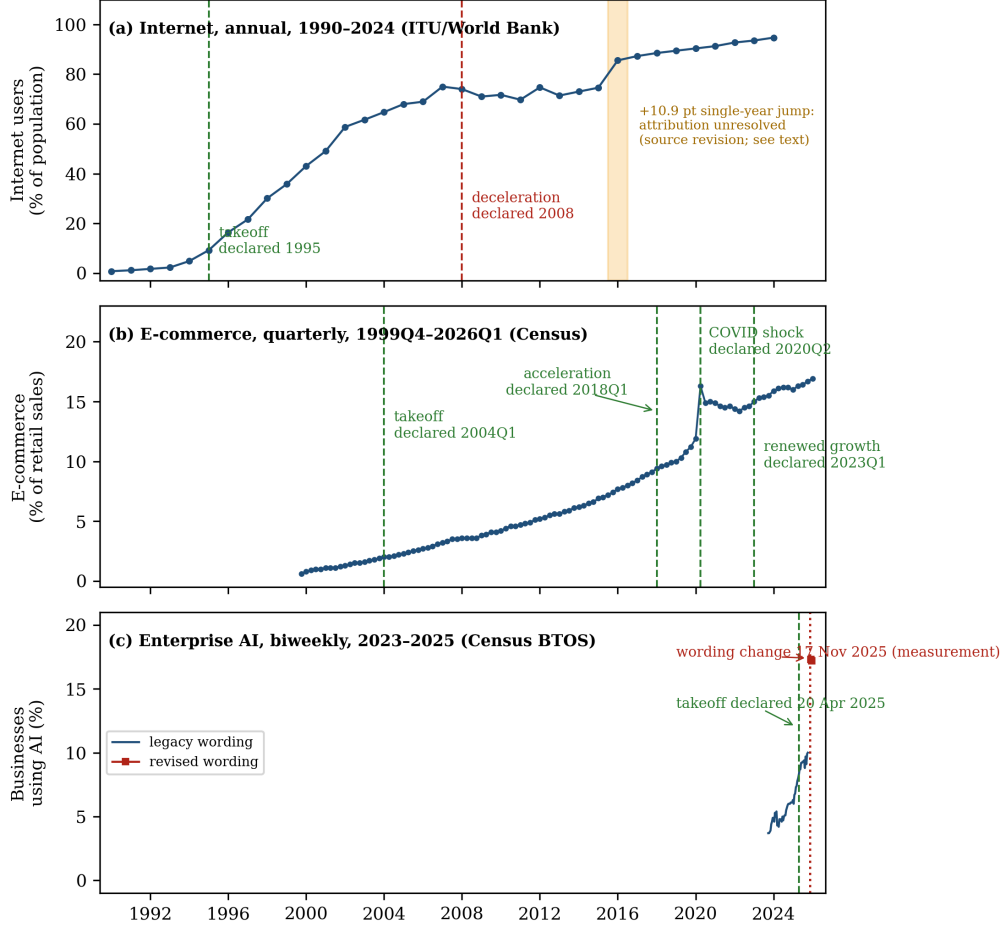


Figure 6: Three decades of U.S. digital technology diffusion under anytime-valid surveillance. (a) Internet users as a share of population, annual (ITU/World Bank via FRED); (b) e-commerce share of retail sales, quarterly (Census via FRED); (c) enterprise AI use, biweekly (Census BTOS; this paper’s core application). Green dashed lines mark declared regime shifts; the red dashed line in (a) marks the declared deceleration; the red dotted line in (c) marks the excluded measurement intervention; the shaded band in (a) marks the 2016 single-year discontinuity whose attribution the available metadata cannot settle. Panels (a) and (b) use the plug-in-variance monitor of Section 3.5; panel (c) uses the exact design-based monitor.

Table 4: Declared regime shifts across three diffusion waves, 1990–2026. The internet and e-commerce rows use the plug-in-variance monitor (approximate validity; Section 3.5); the enterprise-AI rows use the exact design-based monitor. Baseline slopes are in percentage points per period of the respective series (year, quarter, biweekly wave). E-values above 10^4 are reported as decisive.

Series and epoch baseline slope	Declared shift	Direction	Terminal E
<i>Internet (annual)</i>			
0 (pre-registered)	1995	up (takeoff)	$> 10^4$
6.75 (anchored 1996–2003)	2008	down (saturation onset)	192.7
1.46 (anchored 2009–2016)	none by 2024	—	0.7
<i>E-commerce (quarterly)</i>			
0 (pre-registered)	2004Q1	up (takeoff)	59.0
0.10 (anchored 2004–2006)	2018Q1	up (acceleration)	52.4
0.32 (anchored 2018–2020)	2020Q2	up (COVID-19 shock)	$> 10^4$
−0.11 (anchored 2020–2022)	2023Q1	up (renewed growth)	79.1
0.13 (anchored 2023–2025)	none by 2026Q1	—	1.3
<i>Enterprise AI (biweekly)</i>			
0 (pre-registered)	20 Apr 2025	up (takeoff)	42.2
0.299 (anchored Jan–Apr 2025)	none by Dec 2025	(deceleration peaked at 24.3)	4.5

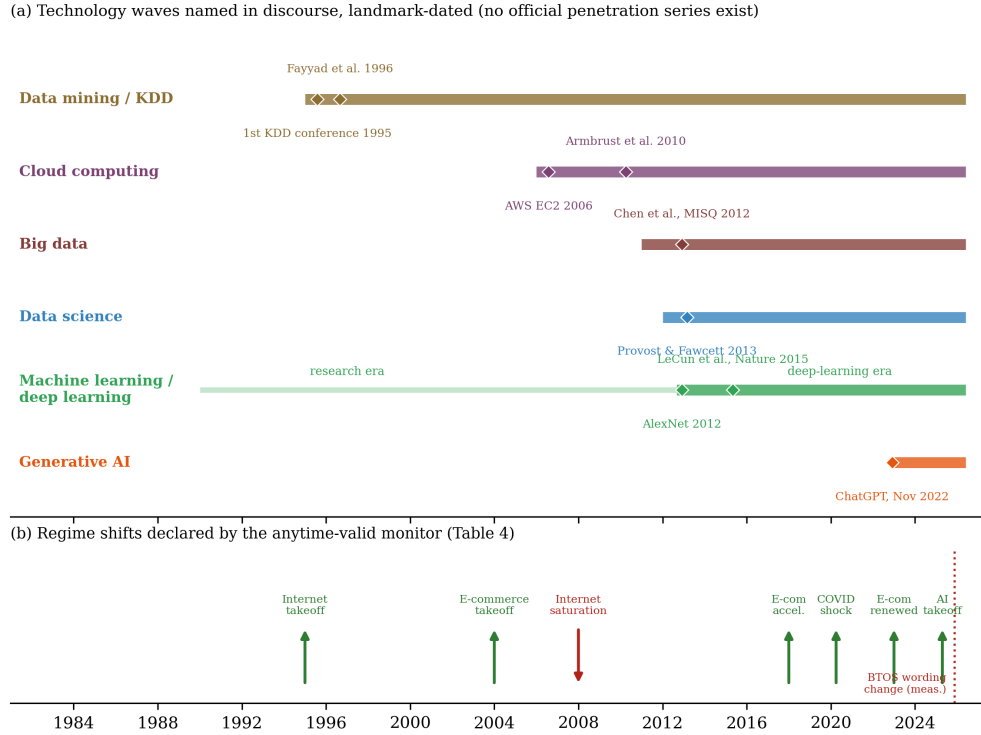


Figure 7: The technology waves of the analytics era, 1990–2026, aligned with the measured record. Panel (a) places the successive waves of technology discourse—data mining and knowledge discovery in databases (Fayyad et al., 1996), cloud computing (Armbrust et al., 2010), big data (Chen et al., 2012), data science (Provost and Fawcett, 2013), the deep-learning breakout within machine learning (Krizhevsky et al., 2017; LeCun et al., 2015), and generative AI—on a common timeline, each dated by canonical, citable landmarks (diamonds) rather than by penetration series, because no official adoption-share statistics were ever produced for these intermediate waves. Panel (b) repeats, on the same axis, the regime shifts declared by the anytime-valid monitor on the three measured series of Table 4. Landmark dates: first ACM KDD conference (1995) and the KDD process framework (Fayyad et al., 1996); Amazon EC2 launch (2006) and the Berkeley view of cloud computing (Armbrust et al., 2010); the MIS Quarterly big-data agenda (Chen et al., 2012); the codification of data science (Provost and Fawcett, 2013); AlexNet (Krizhevsky et al., 2017) and the Nature review of deep learning (LeCun et al., 2015); and the public release of ChatGPT (November 2022), the proximate trigger of the generative-AI wave (Bick et al., 2025).

614 6. Discussion

615 6.1. *Implications for technology and innovation management research*

616 Methodologically, the results argue that the arrival of streaming adoption
617 data requires a matching change in inferential practice. Any TIM study that
618 monitors an ongoing series—of adoption, patenting, platform activity—and
619 reports significance at a data-dependent stopping point inherits the peeking
620 problem, and Table 2 shows how badly it bites on real official data: even Bon-
621 ferroni correction leaves a rule that alarms on nearly two-thirds of looks and
622 certifies a survey redesign as the largest diffusion event on record. Anytime-
623 valid tools remove the problem at negligible implementation cost—equation
624 (3) is one line—and change what a “finding” is: instead of a curve inflec-
625 tion estimated in hindsight, the analyst obtains a dated, auditable evidence
626 trajectory recording exactly how much support a regime shift had at every
627 moment. For the evolutionary perspective animating this special issue, that
628 trajectory is the natural object: takeoffs, stalls, and punctuations become
629 detected events with attached error guarantees, and the 2023–2025 history
630 of enterprise AI—spurt, stall, sustained acceleration, plateau, resumption—is
631 recovered here as precisely such a sequence.

632 Theoretically, treating measurement redesigns as causal interventions re-
633 frames the survey-comparability debate. The dispersion of AI adoption esti-
634 mates across instruments (Crane et al., 2025) and the near-doubling of the
635 BTOS measure under broader wording (Bick, 2026) are usually offered as
636 reasons to distrust any single series. In our framework they are identified
637 components of the observation function g in (1), and the November 2025
638 episode becomes a validation asset: a monitor that cannot absorb a known

instrument intervention should not be trusted to interpret unknown ones. The cross-strata signature developed here—synchronization, extremity relative to each stratum’s own history, mechanical sign coherence—gives applied researchers a portable screening rule for the redesigns that high-frequency programs will keep producing.

6.2. *Implications for practice*

For managers, the immediate application is internal. Enterprise AI deployments generate adoption telemetry—weekly active users, function-level usage, ticket volumes—that is reviewed continuously and acted upon under exactly the conditions that invalidate conventional tests. The protocol of Table 1 transfers directly: define the baseline path for a pilot, run upward and downward monitors at a chosen α , treat tool migrations and instrumentation changes as measurement interventions, and tie scale/hold/stop decisions to evidence thresholds. The Epoch-2 episode is the template for the hardest managerial call: when a promising deployment appears to stall, the monitor distinguishes “24-to-1 evidence of a momentum break—prepare contingencies” from “regime change certified—act,” and in this instance the distinction saved the (hypothetical) decision-maker from killing a trajectory that resumed within weeks. For analysts and policymakers consuming official statistics, the framework supplies a disciplined answer to the recurring question of whether the latest print signals genuine change, with error properties invariant to how often the question is asked.

661 *6.3. Limitations and future research*

662 Five limitations bound the claims. First, the public BTOS tabulations are
663 aggregates from a rotating panel: the monitor tracks population-level rates,
664 and firm-level regime dynamics—entry into and exit from AI use—are observ-
665 able only through restricted microdata access, a natural Federal Statistical
666 Research Data Center extension. Second, the adoption measure is a binary
667 self-report; supplements measuring use across business functions (U.S. Cen-
668 sus Bureau, 2026b) invite multivariate e-processes over adoption depth, not
669 just incidence. Third, the sharp null attributes all within-regime dispersion to
670 design-based sampling error; the κ -robustness analysis shows detection tim-
671 ing depends materially on that attribution, and developing e-processes with
672 data-driven yet valid overdispersion allowance is an open problem. Fourth,
673 the attribution diagnostic for unannounced breaks is a screening rule with
674 a causal rationale rather than an identified estimator; formal identification
675 under richer instrument models would connect this work more deeply to the
676 invariance literature. Fifth, the exact guarantees cover one technology in one
677 country through December 2025, with only two post-redesign waves—too few
678 to re-anchor the revised-regime monitor within sample; the 1990–2026 ret-
679 rospective extends the surveillance logic across technologies but only with
680 approximate, plug-in-variance validity and at coarse temporal resolution, so
681 replications on the extending BTOS series, on the Eurostat enterprise ICT
682 surveys, and on other economies’ high-frequency programs remain the path
683 to a comparative evolutionary evidence base with exact error control.

684 7. Conclusion

685 Technology adoption is now measured as a stream, but inference about
686 it is still largely conducted as if data collection ended yesterday. This pa-
687 per contributes a monitoring framework in which evidence about diffusion
688 regime shifts is anytime-valid—robust to continuous inspection and optional
689 stopping—and causally insulated from changes in the measurement appa-
690 ratus itself. On the biweekly U.S. Census series for enterprise AI use, the
691 framework declared the diffusion takeoff on 20 April 2025 at 42-to-1 evidence,
692 held the mid-2025 deceleration at a peak of 24-to-1 without a false decla-
693 ration, and absorbed a 26-standard-error survey redesign that conventional
694 tests certify as the largest diffusion shock on record. Extended retrospectively
695 across three decades, the same logic recovers the evolutionary record of U.S.
696 digital diffusion—the internet’s 1995 takeoff and 2008 saturation onset, e-
697 commerce’s 2004 takeoff, 2018 acceleration, 2020 pandemic shock, and 2023
698 re-acceleration—as a sequence of dated, directionally signed declarations,
699 with the 2016 internet-series discontinuity standing as a reminder of what
700 is lost when measurement travels without metadata. The broader proposal
701 to the field is a change of stance: from fitting diffusion curves after the fact
702 to running always-on, error-controlled surveillance of how social technologies
703 evolve through the economy—and acting on that evidence with guarantees
704 that survive the way evidence is actually consumed.

705 Declaration of competing interest

706 The authors declare that they have no known competing financial inter-
707 ests or personal relationships that could have appeared to influence the work

708 reported in this paper.

709 Data availability

710 All data are public U.S. Census Bureau Business Trends and Outlook Sur-
711 vey statistics (https://www.census.gov/hfp/btos/data_downloads). The
712 long-run series are retrieved from FRED: Internet users for the United States
713 (ITNETUSERP2USA; <https://fred.stlouisfed.org/series/ITNETUSERP2USA>)
714 and E-Commerce Retail Sales as a Percent of Total Sales (ECOMPCTSA;
715 <https://fred.stlouisfed.org/series/ECOMPCTSA>). The exact analysis
716 datasets (four CSV files: the spliced BTOS national and employment-size-
717 class series with design-based standard errors and vintage provenance, and
718 the two long-run series as retrieved) and replication scripts reproducing ev-
719 ery number, table, and figure in the paper are provided as supplementary
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724 Appendix A. The one-sided mixture e-process

725 *Derivation of (3).* Under $H_0^{(e)}$ in (2), the standardized increments $z_1, z_2, \dots$
726 are i.i.d. $\mathcal{N}(0, 1)$. For a fixed drift δ , the likelihood ratio of $\mathcal{N}(\delta, 1)$ against
727 $\mathcal{N}(0, 1)$ after n observations is $L_n(\delta) = \exp(\delta S_n - n\delta^2/2)$ with $S_n = \sum_{s \leq n} z_s$,
728 a nonnegative martingale with $L_0 = 1$. Mixing over the half-normal prior

729 $\pi_\tau(\delta) = \frac{2}{\tau}\varphi(\delta/\tau)\mathbf{1}\{\delta > 0\}$ preserves the martingale property (Fubini), giving

730 $E_n^+ = \int_0^\infty L_n(\delta) \pi_\tau(\delta) d\delta$. Completing the square with $a_n = n + \tau^{-2}$,

$$E_n^+ = \sqrt{\frac{2}{\pi\tau^2}} \int_0^\infty \exp\left(-\frac{a_n}{2}\left(\delta - \frac{S_n}{a_n}\right)^2 + \frac{S_n^2}{2a_n}\right) d\delta = \frac{2}{\tau\sqrt{a_n}} \Phi\left(\frac{S_n}{\sqrt{a_n}}\right) \exp\left(\frac{S_n^2}{2a_n}\right),$$

731 which is (3); at $n = 0$, $a_0 = \tau^{-2}$ and $E_0^+ = 2\Phi(0) = 1$.

732 *Validity under the composite one-sided null..* Suppose more generally that

733 $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\mu_t, 1)$ with $\mu_t \leq 0$ (drift no greater than the baseline). For

734 any $\delta > 0$, $\mathbb{E}[e^{\delta z_t - \delta^2/2} \mid \mathcal{F}_{t-1}] = e^{\delta\mu_t} \leq 1$, so each $L_n(\delta)$ is a nonnegative

735 supermartingale, and hence so is the mixture E_n^+ . Ville's inequality for non-

736 negative supermartingales then gives $\mathbb{P}(\exists n : E_n^+ \geq 1/\alpha') \leq \alpha'$, establishing

737 (4) with $\alpha' = \alpha/2$; the same argument applies to E_n^- under $\mu_t \geq 0$, and

738 running both monitors controls the per-epoch false-declaration probability

739 at α by a union bound.

740 Appendix B. Implementation parameters

741 All computations use only information available at each wave, preserving

742 predictability. Defaults: prior scale $\tau = 1$; level $\alpha = 0.05$ split equally across

743 the two one-sided monitors (declaration threshold 40); anchoring window

744 $w = 8$ waves with weighted-least-squares slope estimation (weights $1/\hat{\sigma}_t^2$) and

745 the slope's standard error added to the increment variance as in (2); Epoch-

746 1 baseline pre-registered at $\beta_1 = 0$; pre-announced instrument-transition

747 increments excluded from all monitors. Comparators: two-sample z -tests

748 on four-versus-four-wave windows using design-based standard errors, un-

749 corrected at the two-sided 5% level or Bonferroni-corrected by the running

750 number of looks; moving-average rule with 3- and 8-wave windows. Retro-
751 spective (plug-in) runs: MAD window $W = 8$; floors $\phi = 0.8$ (internet, per
752 year) and $\phi = 0.08$ (e-commerce, per quarter); residuals seeded from each an-
753 choring window; e-values capped at 10^4 for reporting. The replication scripts
754 (`reproduce_results.py` for the BTOS application; `reproduce_retro.py` for
755 the 1990–2026 retrospective) reproduce all reported values from the supple-
756 mentary CSV files.

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